

Technical Document #799: Cloud Computing - Comprehensive Overview

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Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Platform as a Service (PaaS) provides a platform for developing and deploying applications.

- Microservices architecture structures applications as a collection of loosely coupled services.
- Kubernetes orchestrates containerized applications across clusters of machines.